Journal of Public Transportation 24 (2022) 100027

Contents lists available at ScienceDirect

Journal of Public Transportation

journal homepage: www.journals.elsevier.com/journal-of-public-transportation

Maximizing access in transit network design

Hema S. Rayaprolu[a,*], Hao Wu[b], Bahman Lahoorpoor[a], David Levinson[a]

[a] School of Civil Engineering, The University of Sydney, Australia
[b] City Futures Research Centre, University of New South Wales, Australia

A R T I C L E I N F O

Keywords:
Transit networks
Optimization
Accessibility
Access-oriented development
Microeconomics

A B S T R A C T

This study adopts an Access-Oriented Design (AOD) framework for optimizing transit network design. We present and demonstrate a method to evaluate the best combination of local and express alternative transit system designs through the novel concept of 'iso-access lines'. Two bus network system designs were explored for a greenfield development in suburban Sydney: through-routed transit lines (T-ways) with higher speeds and more direct service, but longer access and egress times, and local routes that provide additional spatial coverage. We developed scenarios with T-ways only, local routes only, and both, and computed transit access to jobs as a cumulative-opportunities measure for each scenario. Local routes offer greater overall access, while T-ways provide greater access-per-unit-cost. The optimal combination of the two was established by generating 'iso-access' lines and determining access-maximizing combinations for a given cost by applying production-theory principles. For 15-min access, the optimal combinations had T-way service frequency equivalent to 0.48 times that of local routes. This ratio increased to 1.45, 2.05 and 2.63 for 30-min, 45- min and 60-min access respectively. In practice, the method can be applied to determine optimal transit combinations for any given budget and desired access level.

Introduction

Planners turn to transit-oriented design (TOD) for new suburban establishments to reduce auto-dependence and related adverse impacts. TODs achieve this by concentrating high density, mixed-use land use around transit stations, thereby increasing access to transit (Dittmar and Ohland, 2012). However, if land-use development does not follow transit infrastructure as planned, which has been found to be the case in many instances, the full potential of TODs is not realized (Deboosere, El-Geneidy and Levinson, 2018). In other words, while TODs optimize access to transit, access to opportunities provided by transit service is overlooked. Access-oriented design (AOD) can overcome this by using access – the ability to reach destinations – as a performance measure to realize the full potential of planned developments (Deboosere, El-Geneidy and Levinson, 2018).

Accessibility, or *access* in short, features as a performance metric in the transport plans of numerous metropolitan regions world-wide (Boisjoly and El-Geneidy, 2017). It is commonly measured in either a gravity-based or a cumulative-opportunities approach. Access measured as cumulative opportunities represents the number of opportunities (destinations) that can be reached by a certain mode within a certain travel time threshold. In the gravity-based approach, closer destinations are given a greater importance and no travel cost constraints are imposed. While gravity-based measures better capture travel behavior, cumulative opportunities are easier to compute and interpret (Boisjoly and El-Geneidy, 2017). In most transport plans, access metrics are nevertheless limited to measuring social equity and environmental justice. Their use as a measure for prioritizing future investments is non-existent. Transit-network design studies generally optimize for travel time measures (Bagloee and Ceder, 2011; Wang et al., 2020), neglecting the interactions with land use.

Access-like performance measures were considered in some evaluation studies. Mamun et al. (2013) proposed a 'Transit Opportunity Index' which combines access to transit services and transit connectivity between origins and destinations. Fu and Xin (2007) proposed a 'Transit Service Indicator' that combines the measures of service frequency, hours of service, route coverage, and travel time. Nevertheless, access measured as the cumulative opportunities reached by a mode within a given time threshold is more systematic and offers greater interpretability (Wu and Levinson, 2020; Levinson and Wu, 2020). Access computations for numerous scenarios are becoming increasingly feasible within reasonable amounts of time thanks to

* Corresponding author.
E-mail addresses: hema.rayaprolu@sydney.edu.au (H. S. Rayaprolu), hao.wu21@unsw.edu.au (H. Wu), bahman.lahoorpoor@sydney.edu.au (B. Lahoorpoor), david.levinson@sydney.edu.au (D. Levinson).

https://doi.org/10.1016/j.jpubtr.2022.100027

Available online xxxx

powerful travel time estimation tools and standardized transit data structures.

In this study, we propose a method to determine the optimal combination of alternative transit systems that optimizes access for a given budget. The method involves generating 'iso-access' lines which indicate all combinations of service provided by the two transit systems that results in the same amount of access, a concept inspired from isoquants of microeconomics' production theory (Varian, 2014). For a given budget and desired level of access, the optimal combination is the one that gives the greatest access for the least cost and represents the trade-off between the two systems. We determine the optimal trade-off by integrating iso-access lines with budget lines which indicate all service combinations of the two systems that can be provided at the same cost. We demonstrate the application of this method with a case study investigating two types of transit systems – through-routed, direct 'Transit ways' (T-ways), and feeder-style local routes – for a greenfield development area in Liverpool, Sydney. T-ways parallel the concept of Bus Rapid Transit (BRT) (Levinson et al., 2002) and are designed as direct, trunk services connecting TOD centers with rail transit on either end. Local routes, on the other hand, provide an additional feeder-style service within TODs thereby providing greater spatial coverage. Consequently, T-ways are cheaper to operate, while local routes offer benefits of greater coverage. Such transit systems have been discussed in previous studies, albeit under different nomenclature. Delbosc et al. (2016) contrast two transit systems – 'mass transit' – direct, frequent services with less spatial coverage, and 'social transit' – indirect, infrequent services with denser spatial coverage. Walker (2008) calls them 'patronage-driven' and 'coverage-driven' services respectively in his presentation of a strategic framework to achieve a trade-off between economic and social goals for transit providers. However, a quantitative exploration of the trade-off between the two systems has not been undertaken.

To summarize, traditional transit network design, even in a TOD-framework, overlooks land use development – a key component to guarantee the utility of the planned transit. An AOD approach addresses this shortcoming by optimizing access to destinations provided by the planned transit. However, access is almost never considered in evaluating transit designs. In this context, we present a method that applies AOD principles to evaluate the optimal combination or the trade-off between transit systems and demonstrate its application through a case study. The remainder of this paper describes the method, and details how this method can be applied in a real-world context.

Method

When evaluating transit systems in an AOD framework, one would investigate the access provided by the different systems, contrast this with the cost of providing service through the different systems, and determine the access-maximizing option for the available budget. When the systems in consideration are complementary in nature, a combination of the systems provides greater access than the individual systems. To evaluate such systems, we introduce the concept of iso-access lines. An iso-access line illustrates the different combinations of the systems in consideration that result in the same amount of access. For two transit systems, X and Y, iso-access lines can be generated, as shown in Fig. 1, like isoquants in production theory by considering the service provided by X and Y as inputs to produce access A as output. For each level of access, iso-access lines are formed by all combinations of X and Y service resulting in that exact amount of access. The greater the combined service, the greater the access. As we go further away from the origin, access represented by iso-access lines increases.

Mathematically, iso-access lines can be modeled with Cobb-Douglas and Translog formulations. The case in our example can be expressed with Cobb-Douglas and Translog models as given in Eqs. (1) and (2) respectively, and can be estimated with Ordinary Least Squares (OLS) estimations.

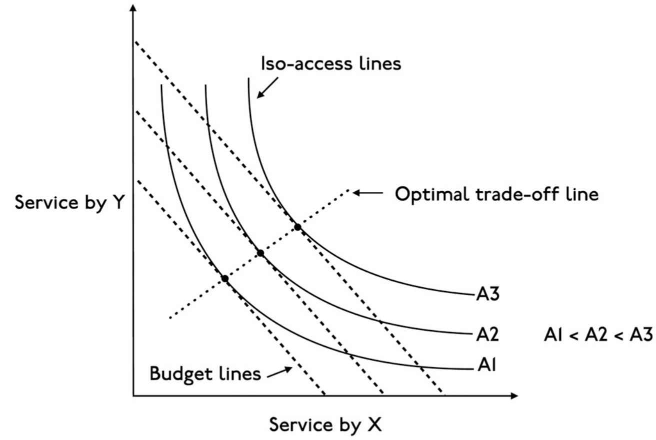

Fig. 1. Solid lines indicate iso-access lines and dashed lines indicate budget lines. Optimal combinations are the points where budget lines are tangent to iso-access lines.

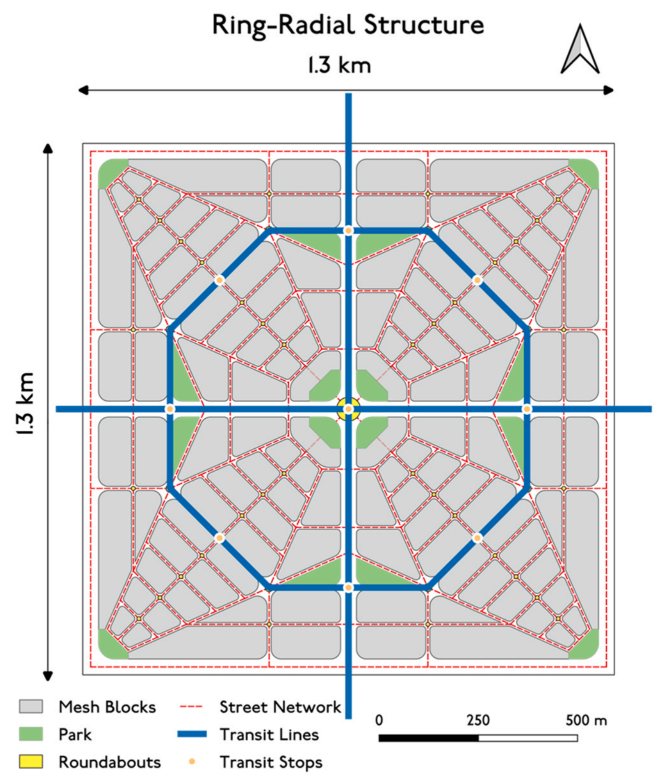

Fig. 2. Superblock layout formed by a ring-radial street network design. Superblocks form the fundamental street network to which transit services and land use are assigned.

$$\ln(A) = I + \beta_X \cdot \ln(X) + \beta_Y \cdot \ln(Y) \quad (1)$$

$$ln(A) = I + \beta_X \cdot ln(X) + \beta_Y \cdot ln(Y) + \beta_{X^2} \cdot ln^2(X) + \beta_{Y^2} \cdot ln^2(Y) + \beta_{XY} \cdot ln(X) \cdot ln(Y) \quad (2)$$

To consider cost constraints, we introduce budget lines, dashed lines in Fig. 1, which are obtained by joining points formed by all combinations of service by X and Y that can be offered at that budget. For a given budget, the optimal combination of service by X and Y that maximizes access is indicated by the point where the budget line corresponding to the given budget is tangent to an iso-access line, and the access indicated by the iso-access line is the maximum access that can be offered by X and Y at the given budget. This is evident from Fig. 1, as at all other points on the budget line, i.e. for all other combinations of X

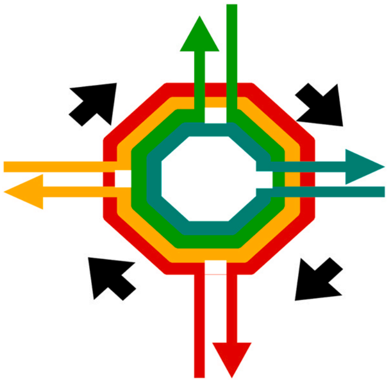

Fig. 3. Stylized local transit octagon. Four local routes, one in each cardinal direction, serve each superblock along the major octagonal road (each color represents a distinct route).

and Y that can be offered at the same cost, access is lower than it is on the iso-access line it is tangent to. Mathematically, these optimum trade-off points can be obtained by determining the points where the slopes of iso-access lines and budget lines are equal.

We call the line joining the so obtained optimal combination points the 'optimal trade-off line'. Any point on the optimal trade-off line gives the access-maximizing combination of X and Y service for a given budget. Alternatively, optimal trade-off lines can be used to determine the cost of providing a level of access with a combination of service by different systems.

Case study

To demonstrate the method described in the previous section, we explored transit options for a greenfield development spanning three peri-urban precincts in the Local Government Area (LGA) of Liverpool within the Greater Sydney region. Currently, these precincts are largely agricultural, and are slated to be developed in the coming decades due to the establishment of the region's second international airport just to the west.

Street network and land use assignment

We assigned street network, land use, and transit services to the development area based on the concept of 'superblocks'. A superblock here, shown in Fig. 2, is a 1300 m x 1300 m symmetric square with two arterials (50–70 km/h) intersecting at the center where the majority of retail and commercial activity is organized around a central green space. The design is our elaboration of Calthorpe's ideal transit-oriented design (Calthorpe, 1993). There are four major roads (30 km/h) on the peripheries. The internal street network is formed by radial streets at 22.5°, 45°, and 67.5° angles from the center, and ring roads connecting the radials. The 22.5° and 67.5° streets bend back to radiate from the outside corners of the superblock, where additional activities might be located. The ring road intersecting the radials at the bend towards the corners is intended as a major octagonal ring road (30 km/h) for automobile traffic circulation. All other internal streets are conceived as shared spaces with a speed of 10 km/h. The street network divides superblocks into smaller blocks and green spaces.

Transit services were routed through the arterials and the major octagonal road with stops at the center and on each face of the major octagonal road. The east-west arterials are intended as trunk transit corridors (70 km/h) as they provide direct and fast connections to the city of Sydney to the east and to the future airport to the west. Private-vehicle traffic from the superblock will exit the streets at the edge of the superblock, and on the north-south arterial (50 km/h), but not the east-west arterial to avoid conflicts with the major transit corridor, and to discourage through travel within the superblock.

We assigned four superblocks to each of the three precincts with the remaining developable land laid out as a 75 m x 150 m grid network, roughly consistent with local development patterns. Land use (population and employment) was allocated to the small street blocks within superblocks in a uniform distribution of residential and employment density. Future population and employment projections, obtained for each precinct as described in Levinson et al. (2020), were divided equally among each of the four superblocks within a precinct. Superblock totals were then distributed across the smaller blocks as a proportion of their area to ensure uniform density.

Transit network layouts

Two transit patterns were explored for connections between the superblocks and to the region's rail network:

- A grid-network of transit ways (T-ways) routed through the north-south and east-west arterials in both directions with rail stations on either end. T-ways stop only at the centers of superblocks they run

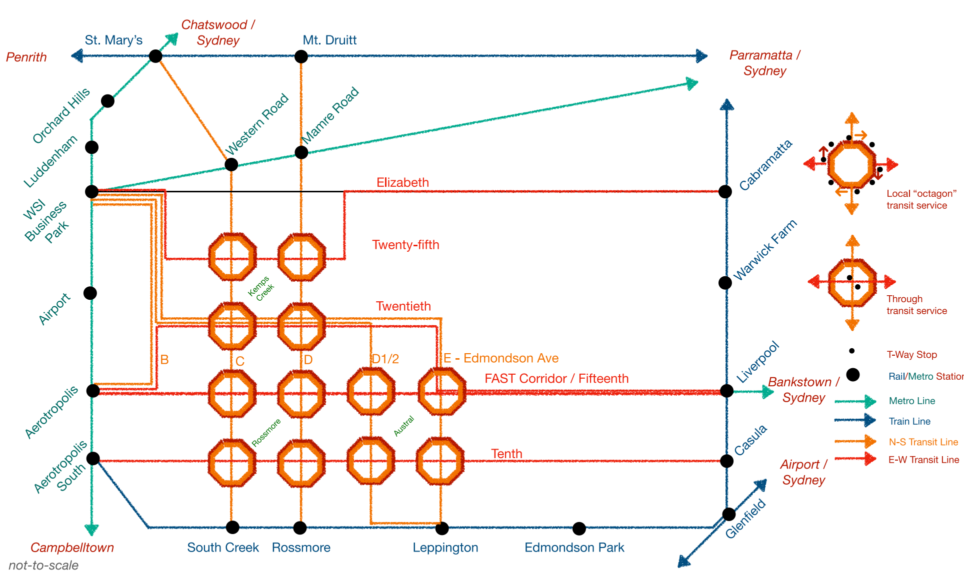

Fig. 4. Schematic representation of transit connections designed for the development area. T-ways connect superblock centers with rail stations on either end. Local routes originate at rail stations, loop around superblocks and terminate at the origin stations.

Table 1
Service levels considered for T-ways and local routes.

Services per hour	Headway (minutes)	T-ways Revenue service hours	Local routes Revenue service hours
1	60	9	35
2	30	18	70
3	20	28	105
4	15	37	140
5	12	47	176
6	10	56	211
8	7	75	281
10	6	94	352
12	5	113	422
15	4	141	528
20	3	188	704
30	2	283	1057
60	1	566	2114

through.

- A set of four local routes per superblock that exit the arterials, run in a clockwise loop along the major octagonal ring road, and then make a round trip back to the station of origin. Local routes do not serve the centers of the superblocks, instead they loop around. They are designed to stop on each face of the major octagonal road, and other stops on the arterials they traverse. Fig. 3 shows how local routes traverse the major octagonal. The octagonal road will have only clockwise service to minimize right-turn conflicts (Australia drives on the left). The downside of such a one-way octagonal loop is illustrated by the case where someone coming from the South and living in the southeast quadrant would have to nearly circumnavigate the whole superblock before exiting. Alternatively, they might exit at the first stop and walk farther. However, from a round-trip perspective, that person has the shortest on-board time when going to the South, as theirs is the last stop before traversing the north-south arterial.

Fig. 4 presents a schematic line diagram of the transit connections envisaged for the development area. T-ways and local routes originate and end at stations on a potential future rail network envisaged for the Greater Sydney region. The future rail network was developed based on the region's future transport plans and is presented in greater detail in Levinson et al. (2020).

T-ways are BRT-style, trunk services stopping at superblock centers and other precinct centers en route to rail stations on either end. Their directness helps maximize temporal frequency. Local routes, on the other hand, serve the same termini and stops as T-ways except within superblocks where they have the advantage of providing greater spatial coverage. We investigated scenarios with T-ways only, with local routes only, and with both together at different services-per-hour. Table 1 lists the 13 different service levels considered. The higher frequencies may appear fiscally infeasible presently, but may become feasible in a future with small, automated buses serving effectively as a fixed-route, on-demand service.

We compared scenarios with T-ways at different service levels to those with local routes at different service levels. Additionally, we built scenarios with both T-ways and local routes provided at all 13 × 13 combinations of services levels, forming a total of 195 scenarios. A set of General Transit Feed Specification (GTFS) files was created for each scenario based on transit route geometries, stop locations and service frequencies. We used QGIS (QGIS Development Team, 2021) to set up the street network and route and stop geometries, and developed a program in Python to generate GTFS files for each scenario.

Access computation

We measured access as cumulative opportunities – the number of opportunities (jobs, people, schools, etc.) that can be reached by a travel mode within a certain amount of time. Our analysis focuses on

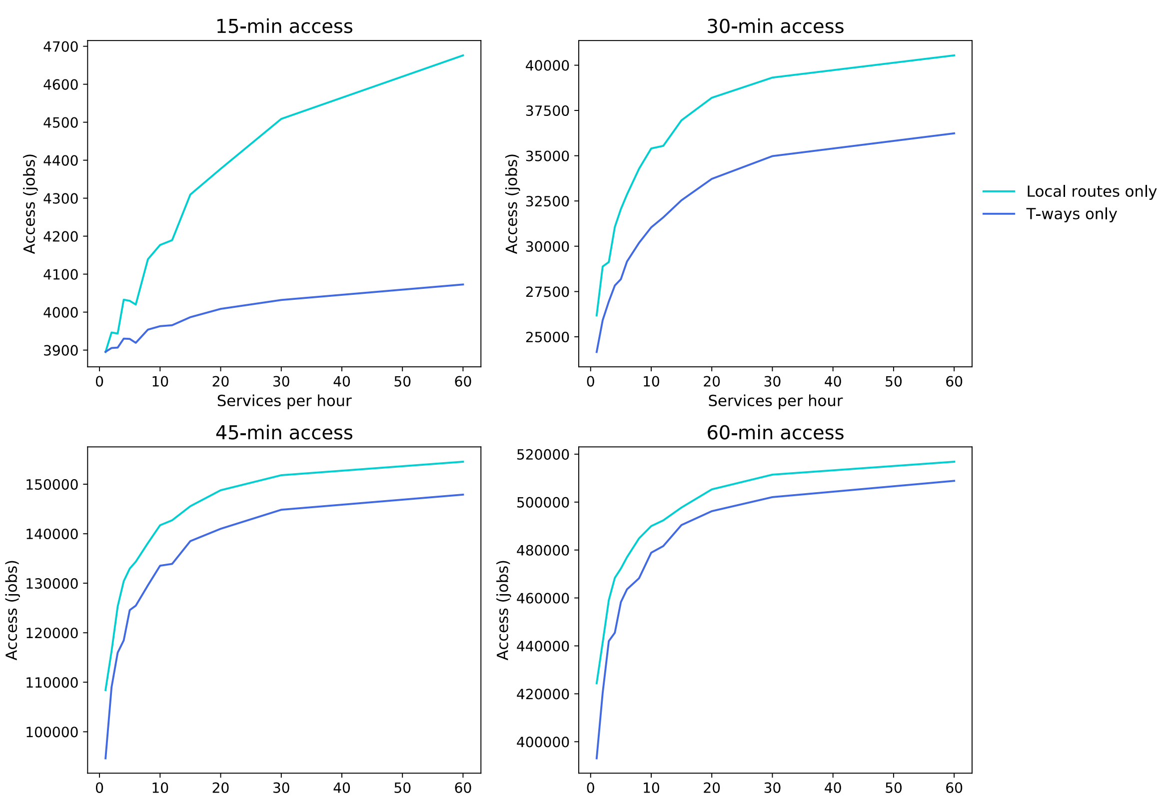

Fig. 5. Comparison of 15-min, 30-min, 45-min and 60-min access provided by T-ways only and local routes only for different service frequencies. Access is higher in scenarios with local routes only than in those with T-ways only across all travel time thresholds.

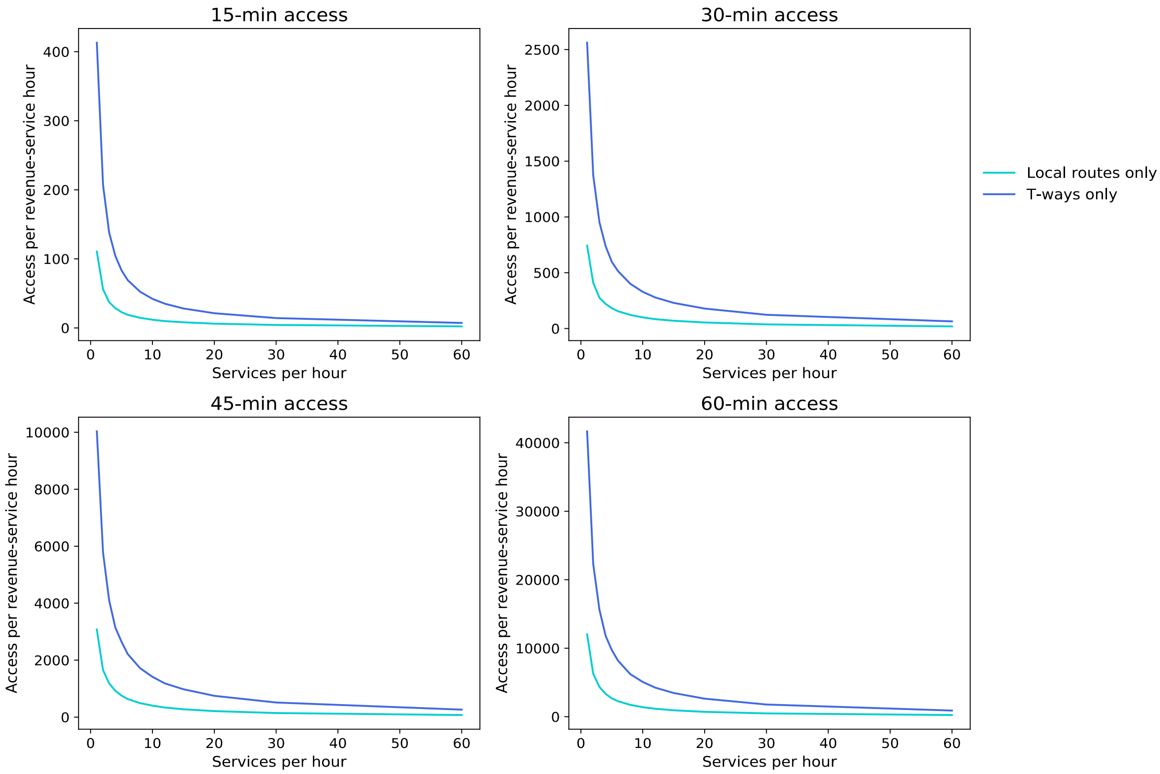

Fig. 6. Comparison of access per unit cost of providing T-ways only and local routes only at different service frequencies. Scenarios with T-ways only offer greater access per unit cost across all travel-time thresholds.

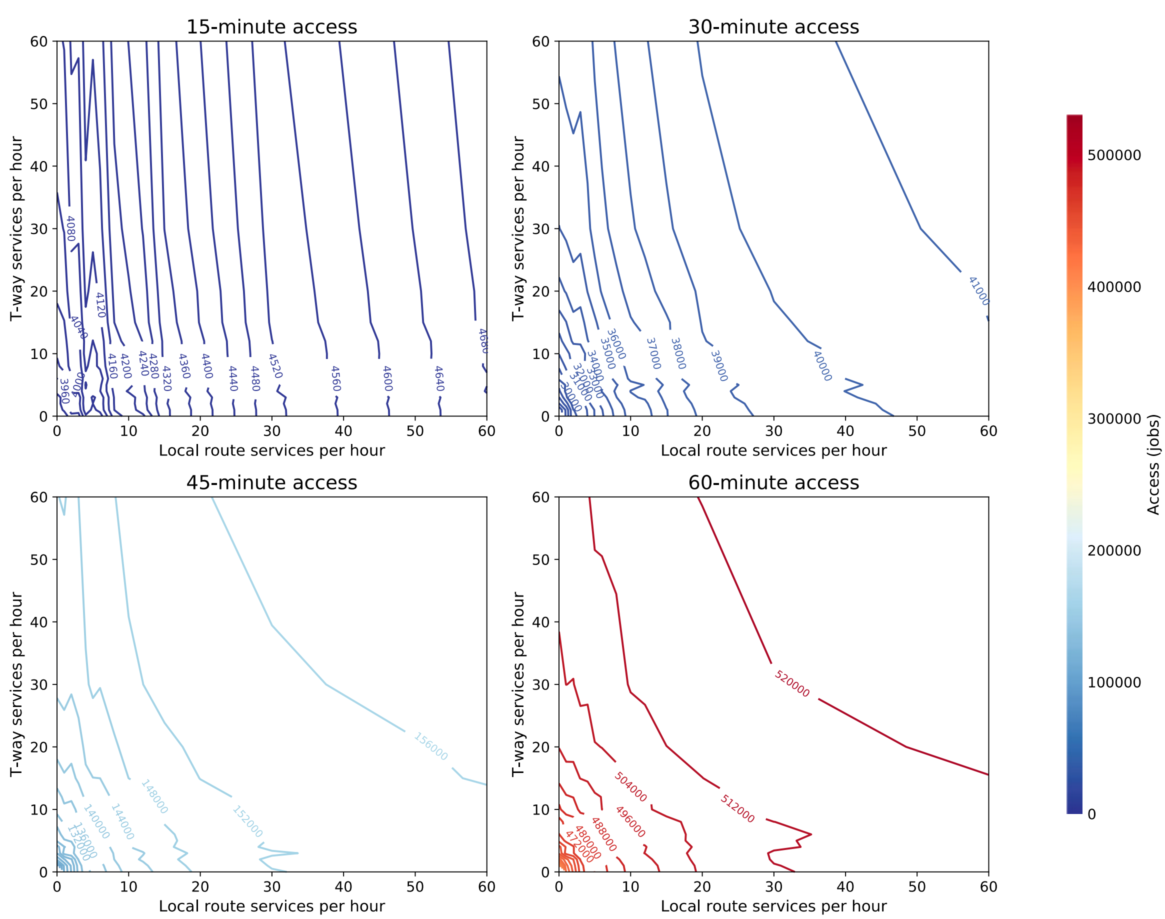

Fig. 7. Observed iso-access lines for 15-min, 30-min, 45-min and 60-min access. The slope of iso-access lines shifts away from local routes and towards T-ways as threshold increases.

Table 2
Model estimates of access: cobb-douglas vs. translog.

	Cobb-Douglas				Translog			
	Coeff	std err	t	P > |t|	Coeff	std err	t	P > |t|
15-min access								
I	8.249	0.003	2606.013	0.000	8.271	0.002	3373.002	0.000
β_T	0.006	0.001	5.904	0.000	0.004	0.002	2.603	0.010
β_L	0.042	0.001	40.796	0.000	0.010	0.002	6.543	0.000
β_{T^2}					0.002	0.000	5.250	0.000
β_{L^2}					0.009	0.000	28.105	0.000
β_{TL}					-0.002	0.000	-6.161	0.000
R^2	0.911				0.986			
30-min access								
I	10.258	0.006	1612.954	0.000	10.187	0.007	1549.889	0.000
β_T	0.033	0.002	15.981	0.000	0.060	0.004	14.196	0.000
β_L	0.076	0.002	36.716	0.000	0.113	0.004	26.840	0.000
β_{T^2}					0.004	0.001	4.232	0.000
β_{L^2}					0.001	0.001	1.217	0.225
β_{TL}					-0.020	0.001	-19.743	0.000
R^2	0.906				0.973			
45-min access								
I	11.716	0.006	2122.758	0.000	11.642	0.006	2111.371	0.000
β_T	0.029	0.002	16.358	0.000	0.058	0.004	16.370	0.000
β_L	0.047	0.002	26.579	0.000	0.091	0.004	25.807	0.000
β_{T^2}					0.002	0.001	2.843	0.005
β_{L^2}					-0.002	0.001	-2.317	0.022
β_{TL}					-0.018	0.001	-20.877	0.000
R^2	0.854				0.961			
60-min access								
I	13.027	0.003	4110.999	0.000	12.984	0.003	4668.318	0.000
β_T	0.019	0.001	18.568	0.000	0.036	0.002	20.278	0.000
β_L	0.024	0.001	23.560	0.000	0.047	0.002	26.710	0.000
β_{T^2}					0.001	0.000	3.445	0.001
β_{L^2}					-0.0003	0.000	-0.860	0.391
β_{TL}					-0.011	0.000	-24.852	0.000
R^2	0.844				0.968			

transit access to jobs within 15, 30, 45 and 60 min.

Using the developed street network and GTFS, we queried travel time isochrones for the approximately 9000 blocks within the Liverpool LGA (comprised by the small blocks within superblocks in the new precincts and the Australian Statistical Geography Standard 'mesh blocks' (ABS, 2022) in the rest of the LGA) for all travel time thresholds in each scenario using Open Trip Planner (OTP, 2022). Population and jobs within a mesh block were assigned to the centroid for computational ease. All queries were run for 8:00 am on a Wednesday. The query time for each scenario was approximately 6 h. The isochrones obtained were used to compute transit access to jobs for each block. The access values computed for mesh blocks were then aggregated as a person-weighted average for the Liverpool LGA.

The person-weighted average access across scenarios were investigated to draw conclusions on the two network systems and their trade-off.

Results and discussion

To compare the two systems individually, we investigated the differences in access, and access-per-unit-cost for scenarios with T-ways only and local routes only. To ascertain the trade-off between the two, we generated iso-access lines and determined optimal trade-off points for different budgets.

T-ways versus local routes

Fig. 5 compares the amount of 15-min, 30-min, 45-min and 60-min access offered for different services-per-hour by local routes only and by T-ways only. Although increasing the number of services-per-hour, in theory, should increase access, some kinks can be observed in the plots, because travel times were estimated at a specific point in time instead of averaging values over a period of time, which would have added orders of magnitude to the computation time.

Scenarios with local routes only provide greater access for all thresholds and at all services-per-hour, highlighting the impact of the wider spatial coverage offered by local routes. The difference was the greatest for 15-min access and decreases as threshold increases, indicating the significance of greater local spatial coverage diminishes as travel time (threshold) increases.

However, T-ways have the advantage of being cost effective due to their direct, non-circuitous alignment. A comparison of the access per unit cost provided by scenarios with local routes only and those with T-ways only is presented in Fig. 6. Cost considered here is indicated by the revenue service hours involved in operating the corresponding service, following the assumption of constant returns to scale (Viton, 1997). The difference in revenue service hours of providing T-ways and local routes for the same amount of service can be seen in Table 1.

As expected, T-ways provided greater access-per-unit-cost across thresholds and services-per-hour. The difference between access-per-unit cost provided by T-ways and by local routes is nearly constant across thresholds.

Trade-off between T-ways and local routes

Given that local routes provide greater access owing to the wider spatial coverage, and that T-ways are more economical, we analyzed the access-maximizing trade-off between the two with the help of iso-access lines.

Fig. 7 shows the iso-access lines for all four thresholds. All service

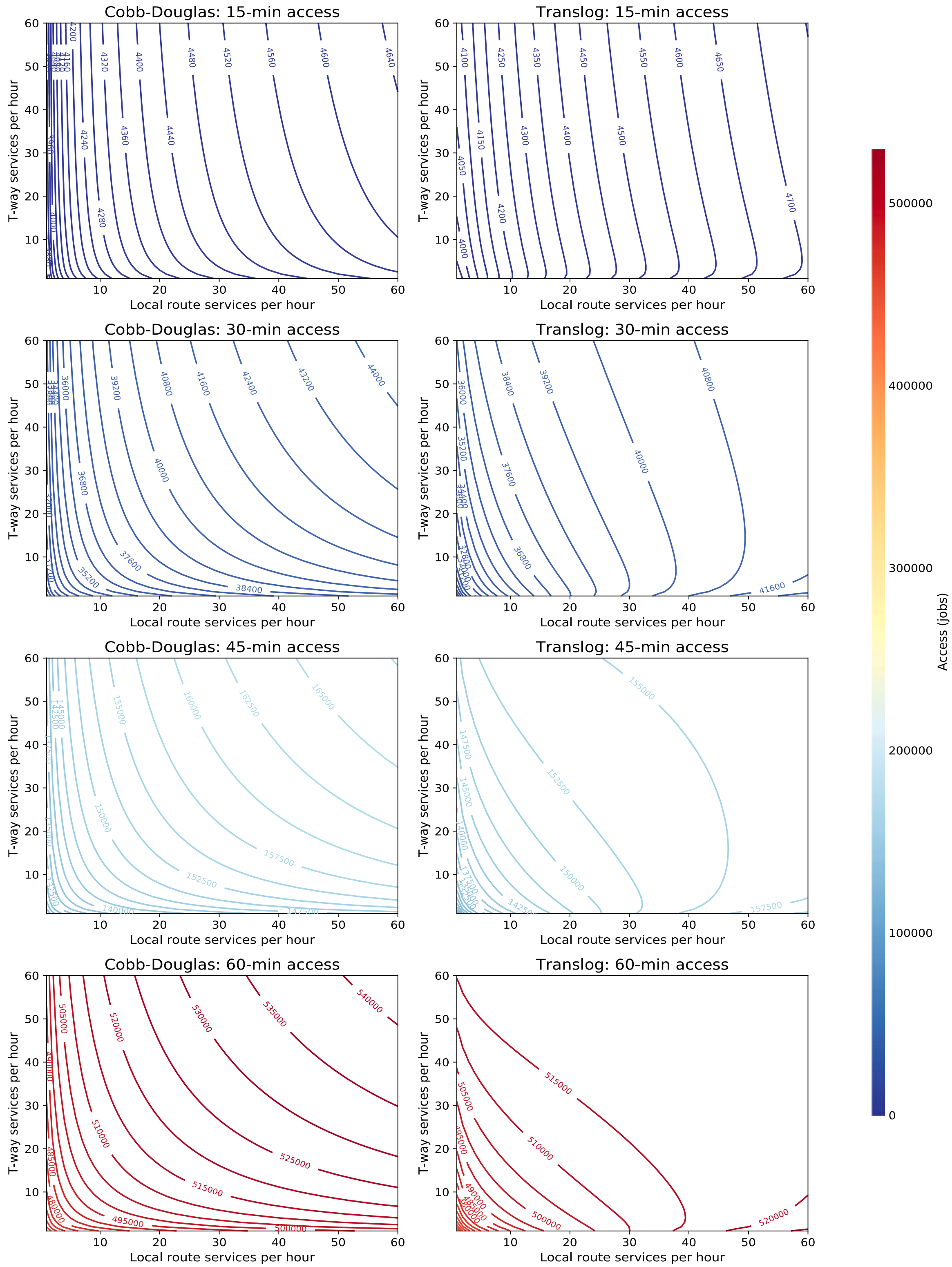

Fig. 8. Comparison of predicted iso-access lines from Cobb-Douglas and Translog models. Predictions from the Translog model do not fit well for service combinations with higher local-route services.

combinations of T-ways and local routes on a line yield the same amount of access. The 15-min iso-access lines are almost vertical indicating local routes provide the major part of access. The lines gradually shift towards T-ways as the threshold is increased.

The plots were developed as contour lines interpolated between the service combinations estimated in the various scenarios. The kinks seen in certain ranges are due perhaps to the gaps in observations and consequent interpolations. To overcome the aberrations, we modeled access and estimated iso-access lines. The estimates obtained on modeling access on T-way and local-route service with Cobb-Douglas and Translog formulations described in Eqs. (1) and (2) are listed in Table 2. The simplicity of the Cobb-Douglas function offers the advantage of greater interpretability, while the Translog function allows us to consider interactions between the two types of services.

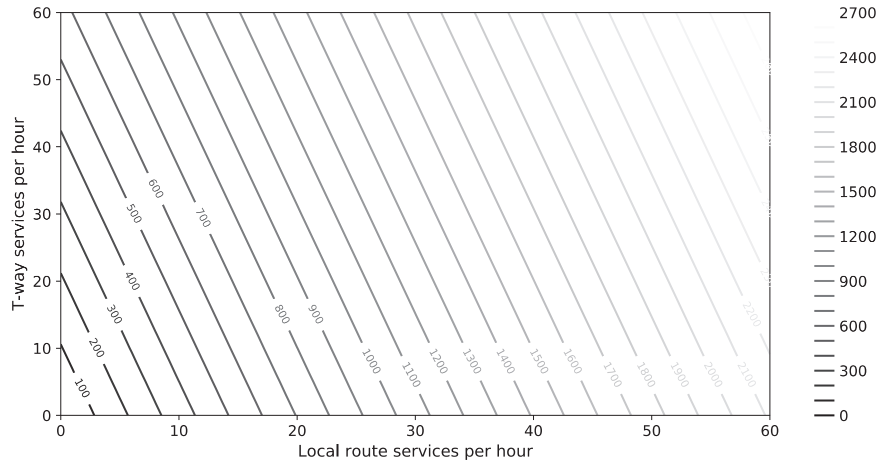

Fig. 9. Budget lines formed by combinations of T-way and local-route service that can be offered at the same cost. Cost is measured as revenue service hours.

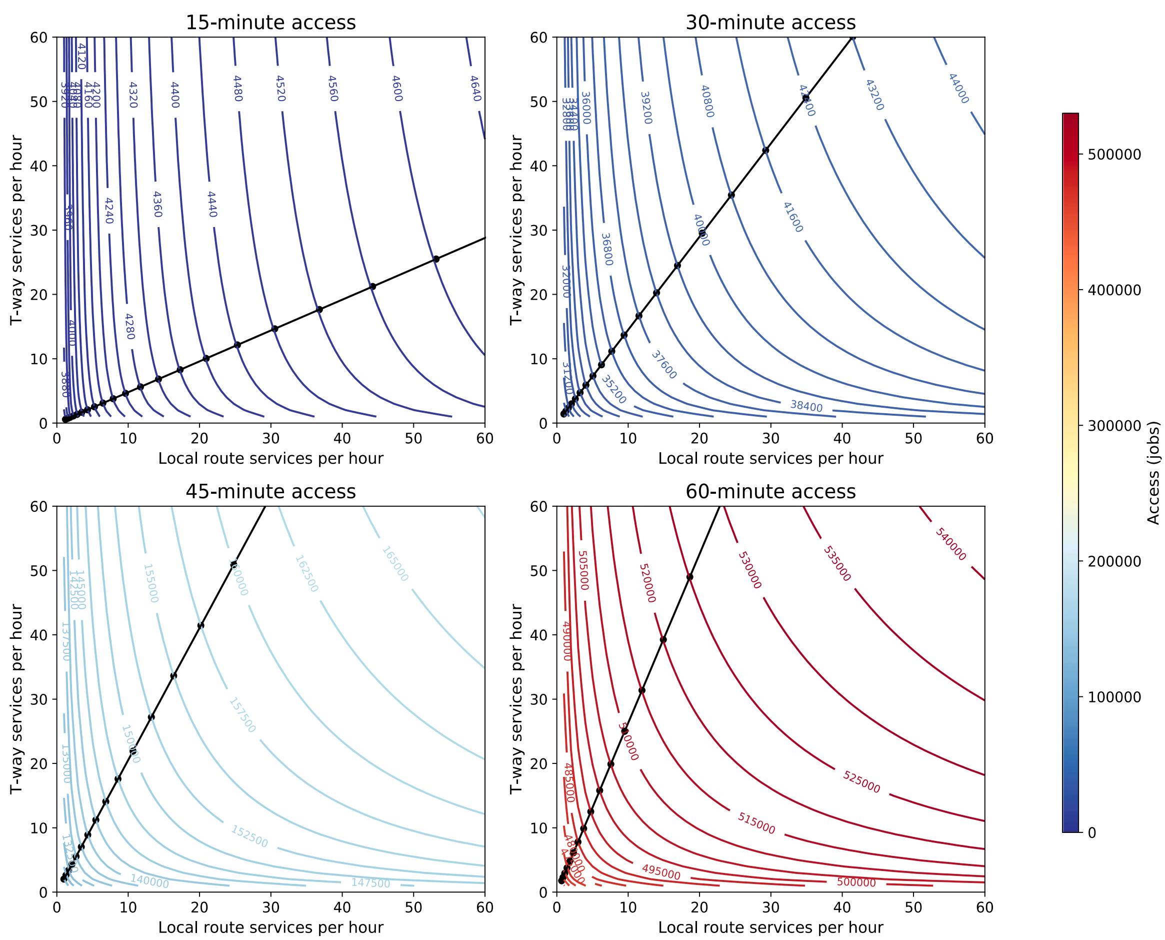

Fig. 10. Optimal trade-off lines showing the optimal combination of T-ways and local routes that maximize access for 15, 30, 45 and 60-minute thresholds. The optimal trade-off line is closer to the local-route service axis for 15-min access and shifts towards T-ways as threshold increases.

In Cobb-Douglas estimations, coefficients represent corresponding elasticities. In our case, a 1% increase in T-way service results in a 0.006% increase in 15-min access, while a 1% increase in local-route service results in a 0.042% increase. Local routes exhibit seven times greater elasticity to 15-min access than T-ways. Local-route elasticity is greater for all access thresholds, however the difference is reduced as threshold increases. For 60-min access, the elasticities are nearly the same. All estimates are highly significant, and the models exhibit a good fit ($R^2 > 0.8$ for all thresholds).

The Translog estimations, on the other hand, demonstrate better model fits ($R^2 > 0.96$). The coefficients of T-way and local-route service (β_T and β_L) exhibit similar trends as the Cobb-Douglas models. The interaction term (β_{TL}) is significant in all models and has a negative coefficient indicating that the influence of T-way service on access decreases as local-route service is increased and vice-versa. Going by the magnitude of the interaction term, the counteracting influence of the services on each other is the lowest for 15-min access, and interestingly drops for 45-min and 60-min access after peaking at 30-min access. Coefficients of the squared terms (β_{T^2} and β_{L^2}) are both positive and significant for 15-min access indicating increasing returns on access on increasing service for both T-ways and local routes individually. The impact for T-ways increases for 30-min access and then decreases for 45-min and 60-min access, and that for local routes decreases as threshold increases and is also not statistically significant in the 30-min, 45-min and 60-min access estimations.

To compare the estimations graphically, we plotted the iso-access lines predicted by the Cobb-Douglas and Translog estimations as shown in Fig. 8. When compared with the observed iso-access lines in Fig. 7, the Translog estimations do not fit well in cases with local-route services greater than 30 services-per-hour, especially for 30-min, 45-min and 60-min access. Consequently, we chose the Cobb-Douglas estimates to determine the optimal trade-off lines.

Optimal trade-off lines were generated by connecting optimal trade-off points – points where the slope of iso-access lines estimated using the Cobb-Douglas estimates shown in Fig. 8 was equal to the slope of budget lines generated as shown in Fig. 9 from T-way and local-route service combinations that have the same cost (revenue-service-hours). The optimal trade-off lines so generated for 15, 30, 45 and 60-min access are presented in Fig. 10.

The slope of the optimal trade-off line for 15-min access is 0.48, indicating the optimal combination to maximize 15-min access should comprise around half as much service by T-ways as by local routes. The slope of the optimal trade-off line for 30-min access is 1.45. Thus, to maximize 30-min access, the optimal service combination bundle should comprise 1.5 times as much service by T-ways as by local routes. The optimal trade-off ratios of T-way service to local-route service for 45-min and 60-min access are 2.05 and 2.63 respectively.

Conclusion

We developed a method to determine the optimal combination of two archetypal transit system designs and demonstrated its application with a case study investigating a through-routed network of direct transit-ways (T-ways) and a system of local routes that provide additional spatial coverage. The method involves generating iso-access lines which depict all combinations of the two systems that yield the same level of access. Applying production theory, we superimpose budget lines onto iso-access lines and determine the points of tangency to generate optimal trade-off lines. Each point on the optimal trade-off lines gives the combination of T-way and local route services that provides the maximum access for the cost involved in providing the services.

We tested two types of bus services in this analysis, however, this method can be applied to establishing the trade-off between any combination of transit alternatives. While the access computation can be time-consuming, depending on the rigor and number of combinations and routes pursued, it can be run in parallel and over multiple machines to speed up the process.

We measured access as cumulative opportunities. Alternatively, gravity access measures can be employed which would generate one optimal trade-off line. However, by using gravity access measures, we would lose visibility of the relative importance of one system over the other for different thresholds (e.g. local vs. long distance trips).

The case study involved multiple assumptions on the transit service layout and land use distribution. Future studies could test the sensitivity and robustness of the resulting optimal combination to these assumptions. One direction could be to vary the land use distribution from uniformly distributed across blocks to a more concentrated pyramid-style distribution with high-rise structures at the center and single-family homes on the fringes of superblocks. Furthermore, our case study does not consider demand for services, which is less accurate than measuring access directly. When demand can be reasonably ascertained, travel time (and thereby access) computations can be modified to consider vehicle capacity, and headways and budgets can also be investigated with greater certainty.

Disciplines

Transportation Engineering.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported as “FAST Corridor Sustainable Urban Mobility Plan” [Project No: 3-010] by City of Liverpool, iMove CRC, and the University of Sydney.

References

ABS, 2022, Mesh Blocks. 〈https://www.abs.gov.au/statistics/standards/australian-statistical-geography-standard-asgs-edition-3/jul2021-jun2026/main-structure-and-greater-capital-city-statistical-areas/mesh-blocks〉.

Bagloee, Saeed Asadi, Ceder, Avishai Avi, 2011. Transit-network design methodology for actual-size road networks. Transportation Research Part B: Methodological 45. Elsevier,, pp. 1787–1804.

Boisjoly, Genevieve, and Ahmed El-Geneidy, 2017, "Measuring Performance: Accessibility Metrics in Metropolitan Regions around the World."

Calthorpe, Peter, 1993. The Next American Metropolis: Ecology, Community, and the American Dream. Princeton Architectural Press,.

Deboosere, Robbin, El-Geneidy, Ahmed M., Levinson, David, 2018. Accessibility-oriented development. Journal of Transport Geography 70. Elsevier,, pp. 11–20.

Delbosc, Alexa, Currie, Graham, Nicholls, Larissa, Maller, Cecily, 2016. Social transit as mass transit in Australian suburban greenfield development. Transportation Research Record 2543. SAGE Publications Sage, Los Angeles, CA, pp. 62–70.

Dittmar, Hank, Ohland, Gloria, 2012. The New Transit Town: Best Practices in Transit-oriented Development. Island Press.

Fu, Liping, Xin, Yaping, 2007. A new performance index for evaluating transit quality of service. J. Public Transp. 10, 4.

Levinson, David, Wu, Hao, 2020. Towards a general theory of access. J. Transp. Land Use (JSTOR) 13, 129–158.

Levinson, David, Hao Wu, Bahman Lahoorpoor, Hema Rayaprolu, Rachel Kohan, and Ben Haddock, 2020, Liverpool Sustainable Urban Mobility Study. Transportlab, University of Sydney. 〈https://ses.library.usyd.edu.au/handle/2123/23732〉.

Levinson, Herbert S., Zimmerman, Samuel, Clinger, Jennifer, Scott Rutherford, G., 2002. Bus rapid transit: An overview. J. Public Transp. 5, 1.

Mamun, Sha A., Lownes, Nicholas E., Osleeb, Jeffrey P., Bertolaccini, Kelly, 2013. A method to define public transit opportunity space. Journal of Transport Geography 28. Elsevier,, pp. 144–154.

OTP, 2022, Intermediate Usage of OpenTripPlanner. 〈http://docs.opentripplanner.org/en/v1.5.0/Intermediate-Tutorial〉.

QGIS Development Team, 2021, "QGIS Geographic Information System." 〈https://www.qgis.org〉.

Varian, Hal R., 2014. Intermediate Microeconomics: A Modern Approach: Ninth International Student Edition. WW Norton & Company.

Viton, Philip A., 1997. Technical efficiency in multi-mode bus transit: A production frontier analysis. Transportation Research Part B: Methodological 31. Elsevier,, pp. 23–39.

Walker, Jarrett, 2008. Purpose-driven public transport: creating a clear conversation about public transport goals. Journal of transport geography 16. Elsevier,, pp. 436–442.

Wang, Chao, Ye, Zhirui, Wang, Wei, 2020. A multi-objective optimization and hybrid heuristic approach for urban bus route network design. *IEEE Access* (IEEE) 8, 12154–12167.

Wu, Hao, Levinson, David, 2020. Unifying access. Transportation Research Part D: Transport and Environment 83 Elsevier,.